

Student project for Ms. Kavya Dave

Degree program: Information and Communication Systems, M.Sc.

Matriculation Number: 565772

Topic

Localization of mobile robots coupling building information modeling and Wi-Fi-based positioning

Lokalisierung mobiler Roboter durch Kopplung von Building Information Modeling und Wi-Fi-basierter Positionierung

Introduction

Mobile robots are increasingly deployed for automated inspection of the built environment. To effectively inspect the built environment, reliable centimeter-level localization of mobile robots using building information modeling (BIM) is crucial. Accurate localization of mobile robots enables consistent navigation, repeatable inspections and measurements, and precise location-referencing of inspection findings. The IEEE 802.11az standard, also known as Next-Generation Positioning, introduces physical-layer features that enhance wireless fidelity (Wi-Fi) ranging and positioning accuracy. The improvements in ranging and positioning accuracy may be utilized for time-of-arrival and angle-of-arrival estimation to provide high resolution and accuracy in robot localization. In this student project, a framework for 3D indoor localization coupling BIM models and Wi-Fi communication standards must be developed. State-of-the-art techniques for 3D indoor positioning using Wi-Fi-based communication are to be reviewed. The framework must be implemented and validated in simulated environments with varying geometries across numerous BIM models.

Tasks

1. Perform a literature review of the state-of-the-art techniques for 3D indoor positioning using Wi-Fi communication with an emphasis on IEEE 802.11az. Analyze state-of-the-art methods, identify challenges, and gaps in the current technology related to BIM-based robot localization.
2. Based on the literature review in Task 1, develop a framework that is capable of 3D indoor positioning using the 802.11az standard for WiFi communication. The framework must be capable of fusing position estimates from robot odometry and WiFi-based positioning.
3. Implement the 3D indoor positioning framework by developing the requisite software modules.
4. Validate the framework implemented in Task 3 by testing indoor positioning of a mobile robot in simulation environments across BIM models.
5. Discuss the results of the validation tests and the localization accuracy of the framework. Discuss accuracy limits, failure cases, and opportunities for future improvement. Provide an outlook on the applicability of the framework in practice.

Assembling the written documentation, also referred to as “the thesis” or “the paper”, will start at the same time when the master thesis period starts, and it will begin with iteratively defining a proper structure together with the supervisor. To ensure transparency and support an open communication during supervision, the documentation will be written in parallel to conducting the tasks defined above; the student will receive further support by discussing the current status of the documentation with the supervisor in regular intervals (at least once a month). Therefore, the current status of the documentation will be submitted to the supervisor at the end of each month (and more frequently when required). To ensure that the thesis can be assessed within the required deadlines, the defense date must be arranged no later than

the midpoint of the thesis work period, but can also be set right at the start of the thesis work, which is solely the responsibility of the student. The thesis must be submitted both digitally (PDF) and as hardcopy (2 copies). The source files of the thesis (Word or LaTeX), the source files of embedded figures and of other related files (sensor data, source codes, etc.) and other sources that are not freely available must be submitted together with the thesis.

Start: 01.09.2025
Submission: 01.03.2026
First Examiner: Professor Dr. Kay Smarsly
Second Examiner: Dr. Alexander Chmelnizkij
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